## exercise 3

① SQL Query

```
SELECT
    product_name,
    price,
    CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000 THEN
        'Mid-range'
    ELSE 'Budget'
    END AS price_category
FROM products;
```

Result

| product_name | price | price_category |
|---|---|---|
| laptop | 1200.00 | Expensive |
| Phone | 800.00 | Mid-range |
| keyboard | 45.00 | Budget |
| Monitor | 300.00 | Mid range |
| Mouse | 25.00 | Budget |

② SQL Query

```
SELECT
    customer_name,
    amount,
    CASE
    WHEN amount >= 1000 THEN 'High Value'
    WHEN amount BETWEEN 500 AND 999.99 THEN
    'Medium Value'

    ELSE 'Low Value'
    END AS order_value_category
    FROM orders;
```

Result

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```
FROM orders;
```

Result

| customer_name | amount | order_value_category |
|---|---|---|
| Alice | 150.00 | low value |
| Bob | 500.00 | Medium value |
| Charlie | 999.99 | Medium Value |
| Diana | 45.50 | Low value |
| Ethan | 1200.00 | High Value |

3.) SQL Query

```
SELECT emp name,
       department,
       salary,
CASE
    WHEN department='IT AND salary >80000 THEN 'Senior IT'
    WHEN department = 'HR' AND salary >55000 THEN Experienced HR'
    ELSE 'Staff'
END AS position_Level
FROM employees;
```

Result

| emp_name | department | salary | position_level |
|---|---|---|---|
| John | IT | 85 000 | Senior IT |
| Sara | HR | 60 000 | experienced HR |
| Mark | IT | 75000 | Staff |
| Lucy | Finance | 95000 | Staff |
| Tom | HR | 55000 | Staff |

4 SQL Query



```
SELECT student_name,
       score,
```

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| | | | |
|---|---|---|---|
| [illegible] | Finance | 95000 | Staff |
| [illegible] | HR | 55000 | Staff |

④ SQL Query

```
SELECT      student_name,
            score,
  CASE
      WHEN score >= 90 THEN 'A'
      WHEN score BETWEEN 80 AND 89 THEN 'B'
      WHEN score BETWEEN 70 AND 79 THEN 'C'
      WHEN score BETWEEN 60 AND 69 THEN 'D'
      ELSE 'F'
  END AS grade
  FROM students;
```

Result

| student_name | score | grade |
|---|---|---|
| Anna | 92 | A |
| Ben | 76 | C |
| Cara | 59 | F |
| David | 83 | B |
| Ella | 68 | D |

5.) SQL Query

```
SELECT  delivery_id,
        delivery_time_minutes,
CASE
      WHEN delivery_time_minutes <= 30 THEN 'Fast'
      WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN
        'On Time'
      ELSE 'Late'
      END AS performance
      FROM deliveries;
```



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END AS performance
FROM deliveries;

Result

| delivery_id | delivery_time_minutes | performance |
|---|---|---|
| 1 | 45 | On Time |
| 2 | 80 | Late |
| 3 | 30 | Fast |
| 4 | 65 | late |
| 5 | 100 | Late |

6.) SQL Query

```
SELECT issue_type,
       priority,
CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

Result

| issue_type | priority | priority_label |
|---|---|---|
| Login issue | 1 | Low |
| Server down | 3 | High |
| Server system | 2 | Medium |
| Email error | 2 | Medium |
| Password reset | 1 | Low |

7.) SQL Query

```
SELECT
    student_id,
    (days_present * 100.0 / total_days) AS attendance_percentage,
CASE
    WHEN (days_present * 100.0 / total_days) >= 90 THEN 'Excellent'
    WHEN (days_present * 100.0 / total_days) BETWEEN 75 AND 89
    THEN 'Good'
```



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```
CASE
    WHEN (days_present * 100.0 / total_days) >= 90 THEN 'Excellent'
    WHEN (days_present * 100.0 / total_days) BETWEEN 75 AND 89
        THEN 'Good'
    ELSE 'Needs Improvement'
END AS attendance_status
FROM attendance;
```

Result

| student_id | attendance_percentage | attendance_status |
|---|---|---|
| 1 | 90% | Excellent |
| 2 | 60% | Needs Improvement |
| 3 | 96% | Excellent |
| 4 | 50% | Needs Improvement |
| 5 | 100% | Excellent |

8.) SQL Query

```
SELECT product_id,
       stock_qty,
CASE
    WHEN stock_qty = 0 THEN 'Out of Stock'
    WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
ELSE 'In Stock'
END AS stock_status
FROM products_inventory;
```

Result

| product_id | stock_qty | stock_status |
|---|---|---|
| 1 | 5 | low Stock |
| 2 | 0 | Out of Stock |
| 3 | 25 | In Stock |
| 4 | 10 | In Stock |
| 5 | 3 | Low Stock |

9.) SQL Query

```
SELECT subject,
       enrolled_students
```

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| | | | |
|---|---|---|---|
| 5 | 3 | Low | Stock |

9) SQL Query

```
SELECT subject,
       enrolled_students,
CASE
    WHEN enrolled_students >= 25 THEN 'Large'
    WHEN enrolled_students BETWEEN 10 AND 24 THEN
        'Medium'
  ELSE 'Small'
END AS class_size_category
FROM classes;
```

Result

| subject | enrolled_students | class_size_category |
|---|---|---|
| Math | 30 | Large |
| English | 25 | Large |
| Science | 15 | Medium |
| Art | 5 | Small |
| History | 20 | Medium |

10) SQL Query

```
SELECT payment_id,
       payment_method
       amount,
CASE
    WHEN payment_method = 'Cash' AND amount >=200 THEN
        'Eligible for Discount'
  ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

Result

| payment_id | payment_method | amount | discount_eligibility |
|---|---|---|---|
| 1 | Card | 500 | Not Eligible |

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| Subject | Enrolled_Students | Class_Size_Category |
|---|---|---|
| Math | 30 | Large |
| English | 25 | Large |
| Science | 15 | Medium |
| Art | 5 | Small |
| History | 20 | Medium |

10) SQL Query

```
SELECT  payment_id,
        payment_method
        amount,
CASE
    WHEN payment_method = 'Cash' AND amount >= 200 THEN
        'Eligible for Discount'
    ELSE 'Not Eligible'
END AS discount_eligibility
FROM payments;
```

Result

| payment_id | payment_method | amount | discount_eligibility |
|---|---|---|---|
| 1 | Card | 500 | Not Eligible |
| 2 | Cash | 200.00 | Eligible for Discount |
| 3 | Card | 150.00 | Not Eligible |
| 4 | PayPal | 75.00 | Not Eligible |
| 5 | Cash | 300.00 | Eligible for Discount |



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